

Advanced Econometrics Lecture 05

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Models for unordered multiple choices

Some studies of multiple-choice settings include the following:

- ① Hensher (1986, 1991), McFadden (1974), and many others have analysed the travel mode of urban commuters.
- ② Schmidt and Strauss (1975a,b) and Boskin (1974) have analysed occupational choice among multiple alternatives.
- ③ Terza (1985a) has studied the assignment of bond ratings to corporate bonds as a choice among multiple alternatives.
- ④ Rossi and Allenby (1999, 2003) studied consumer brand choices in a repeated choice (panel data) model.
- ⑤ Train (2003) studied the choice of electricity supplier by a sample of California electricity consumers.
- ⑥ Hensher, Rose, and Greene (2006) analysed choices of automobile models by a sample of consumers offered a hypothetical menu of features.

Source: W.H. Greene, *Econometric Analysis*, Sixth Edition, p. 841.

Unordered choice models

- ① Multinomial logit model
 - Independence of Irrelevant Assumption (IIA)
- ② Multinomial probit model
 - Computation is used to be extremely time-consuming. (Greene (2008) p. 850)
- ③ Nested logit model
- ④ Mixed logit model
- ⑤ Conditional logit
- ⑥ Other models

Basic unordered choice models

1 Multinomial logit

- Explanatory variables are individual-specific; they do not change across alternatives.
- Base alternative
- The Independence from Irrelevant Alternatives Assumption (IIA).
- For example marital status determinants: married/cohabitant, widow/widower, divorced, separated, bachelor.

2 Conditional logit

- Explanatory variables are individual and alternative specific; they vary from individual to individual and are different for each choice the consumer might make.
- In conditional logit besides entities' characteristics, we have to have alternatives' characteristics.
- For example choice of fishing mode: beach, pier, private boat, charter boat, while we have the price of each alternative mode, catch rate per alternative, **income**.

Multinomial logit - Greene p. 843-844

The multinomial logit model is

$$Prob(Y_i = j|\mathbf{x}) = \frac{\exp(\mathbf{x}'_i\beta_j)}{\sum_{j=0}^J \exp(\mathbf{x}'_i\beta_j)}, \quad j = 0, 1, 2, \dots, J. \quad (1)$$

A convenient normalization that solves the problem [of indeterminacy] is $\alpha_0 = \mathbf{0}$.

This arises because the probabilities sum to one, so only J parameter vectors are needed to determine the $J + 1$ probabilities.

Therefore, the probabilities are

$$Prob(Y_i = j|\mathbf{x}) = \frac{\exp(\mathbf{x}'_i\beta_j)}{\sum_{k=0}^J \exp(\mathbf{x}'_i\beta_k)}, \quad (2)$$

where $k = 0, 1, 2, \dots, J$, and $\beta_0 = \mathbf{0}$.

Fishing modes explained with individual specific vars only

variable name	variable label
mode	Fishing mode
price	price for chosen alternative
crate	catch rate for chosen alternative
pbeach	price for beach model
ppier	price for pier mode
pprivate	price for private boat mode
pcharter	price for charter boat mode
qbeach	catch rate for beach mode
qpier	catch rate for pier mode
qprivate	catch rate for private boat mode
qcharter	catch rate for charter boat mode
income	monthly income in thousands \$

Data comes from A. Colin Cameron and Pravin K. Trivedi, Microeconometrics Using Stata, Revised Edition, Stata Press, 2010.

Multinomial logit – R output

```
> summary(mlogit1)
```

Call:

```
mlogit(formula = mode ~ 0 | income, data = Fish, method = "nr")
```

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)	
(Intercept):boat	0.738921	0.196731	3.7560	0.0001727	***
(Intercept):charter	1.341291	0.194517	6.8955	0.0000000000005367	***
(Intercept):pier	0.814150	0.228632	3.5610	0.0003695	***
income:boat	0.091906	0.040664	2.2602	0.0238116	*
income:charter	-0.031640	0.041846	-0.7561	0.4495908	
income:pier	-0.143403	0.053288	-2.6911	0.0071223	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -1477.2

McFadden R²: 0.013736

Likelihood ratio test : chisq = 41.145 (p.value = 0.0000000060931)

Coefficient interpretation

- A.C. Cameron, P.K. Trivedi, *Microeconometrics Using Stata*, p. 500
- Coefficients in a multinomial model can be interpreted in the same way as binary logit model parameters are interpreted, with comparison being to the base category.
- So a positive coefficient from `mlogit` means that as the regressor increases, we are more likely to choose alternative j than alternative 1. This interpretation will vary with the base category and is clearly most useful when there is a natural base category.

Multinomial logit – marginal effects for means

```
> # compute a data.frame containing the mean value
> # of the covariates in the sample
> z <- with(Fish, data.frame(income =
                                tapply(income, index(mlogit1)$alt, mean)))

> print(z)
      income
beach  4.099337
boat   4.099337
charter 4.099337
pier   4.099337
>
> # compute the marginal effects for average characteristics
> # impact of an additional thousand of dollars
> effects(mlogit1, covariate = "income", data = z)
      beach      boat      charter      pier
0.00007495795  0.03259851598 -0.01201366829 -0.02065980565
```

Multinomial logit – AME Average Marginal Effects

```
# compute the average marginal effects
AME = matrix(0, nrow=1182, ncol=4)
# nrow = nrow(Fishing) # Fishing not Fish
# ncol = the number of alternatives

for(iter in 1:1182) {
  income = Fish$income[Fish$chid==iter]
  z = data.frame(income =
    tapply(income, index(mlogit1)$alt[Fish$chid==iter], mean))
  AME[iter, ] = effects(mlogit1, covariate = "income", data = z)
}

AvMargEff = colMeans(x=AME)
print(AvMargEff)
> print(AvMargEff)
[1] 0.0001646645 0.0317561616 -0.0111518637 -0.0207689624
```

Independence of Irrelevant Alternatives

- P. Ray, Independence of Irrelevant Alternatives, *Econometrica*, Vol. 41, No. 5 (Sep., 1973), pp. 987-991
- THE CONDITION OF "independence of irrelevant alternatives" was first used by Arrow in 1951. (Arrow, K. J.: Social Choice and Individual Values. 2nd Ed. New Haven: Yale University Press, 1963)
- Arrow defines independence of irrelevant alternatives as follows: IIA(A):
"Let $R_1, \dots, R_n$ and $R'_1, \dots, R'_n$ be two sets of individual orderings and let $C(S)$ and $C'(S)$ be the corresponding social choice functions. If, for all individuals i and all x and y in a given environment S , xR_iy if and only if xR'_iy , then $C(S)$ and $C'(S)$ are the same.
- In discrete choice theory, the IIA assumption says that when people are asked to choose among a set of alternatives, their odds of choosing A over B should not depend on whether some other alternative C is present or absent. Source: <https://statisticalhorizons.com/iia>

Independence of Irrelevant Alternatives

- When I'm ordering ice cream, the relative probability that I choose chocolate chip over vanilla isn't affected by whether strawberry is an option.
- I prefer listening to Stravinsky over Madonna. Even if Spotify suggests I listen to Justin Bieber, I will still prefer Stravinsky over Madonna.
- Source: <https://www.quora.com/Whats-an-intuitive-explanation-of-Independence-of-Irrelevant-Alternatives>

Independence from irrelevant alternatives

Does alternative "pier" violate IIA assumption?

```
> # independence from irrelevant alternatives assumption
> mlogit1 = mlogit(mode ~ 0 | income, data = Fish,
                  refllevel="beach")
> mlogit2 = mlogit(mode ~ 0 | income, data = Fish,
                  refllevel="beach",
                  alt.subset=c("beach", "boat", "charter"))
> # compute the test
> hmf test(mlogit1, mlogit2)
```

Hausman-McFadden test

```
data: Fish
chisq = 4.6, df = 4, p-value = 0.3
alternative hypothesis: IIA is rejected
```

Combining alternatives

Let's test the hypothesis that private boat and charter boat alternatives might be combined into one category

i.e. non-constant variables' parameters in these categories are equal to themselves.

$$\beta_{income}^{boat} = \beta_{income}^{charter}$$

```
> linearHypothesis(model=mlogit1, c("income:boat=income:charter"))
```

Linear hypothesis test

Hypothesis:

income:boat - income:charter = 0

Model 1: restricted model

Model 2: mode ~ 0 | income

	Res.Df	Df	Chisq	Pr(>Chisq)
1	1177			
2	1176	1	19.594	0.000009577 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

A note on multinomial probit

Greene p. 850-851

A natural alternative model that relaxes the independence restrictions built into the multinomial logit (MNL) model is **multinomial probit model** (MNP). The structural equations of the MNP model are

$$U_{ij} = \mathbf{x}'_{ij}\boldsymbol{\beta} + \varepsilon_{ij}, \quad j = 1, 2, \dots, J, \quad [\varepsilon_{i1}, \varepsilon_{i2}, \dots, \varepsilon_{iJ}] \sim N(\mathbf{0}, \Sigma) \quad (3)$$

For more than two choices, this specification is far more general than the MNL model, which assumes $\Sigma = I$.

The main obstacle to implementation of the MNP model has been the difficulty in computing the multivariate normal probabilities for any dimensionality higher than 2.

Multinomial logit models in R

① multinom – library nnet

- Easier to estimate

② mlogit – library mlogit

- More challenging
- Data preparation
- Marginal effects
- `hmfetest` function

Employer from which person received the most earnings

variable name	variable label
empltyp1	Employer from which reference person received the most earnings in past 12 months
age_ref	Age of reference person
sex_ref	Sex of reference person

No	Label
1	Private company
2	Federal government
3	State government
4	Local government
5	Self-employed

Data comes from Consumer Expenditure Surveys, Bureau of Labor Statistics, United States Department of Labor

multinom – library nnet – data form

id	age_ref	emplytyp1	famsize	marital1	sex_ref
1	54	Private company	2	3	2
2	29	Private company	3	1	2
3	51	State government	2	1	2
4	29	Private company	4	1	2
5	64	Private company	3	1	2
6	35	Private company	2	5	2
7	22	State government	2	5	2
8	44	Private company	1	5	1
9	51	Federal government	1	3	1

mlogit – library mlogit – data form

	age	emplyt1	famsize	marital	sex	chid	alt
1.Federal government	54	FALSE	2	3	2	1	Federal government
1.Local government	54	FALSE	2	3	2	1	Local government
1.Private company	54	TRUE	2	3	2	1	Private company
1.Self-employed	54	FALSE	2	3	2	1	Self-employed
1.State government	54	FALSE	2	3	2	1	State government
2.Federal government	21	FALSE	2	1	1	3	Federal government
2.Local government	21	FALSE	2	1	1	3	Local government
2.Private company	21	TRUE	2	1	1	3	Private company
2.Self-employed	21	FALSE	2	1	1	3	Self-employed
2.State government	21	FALSE	2	1	1	3	State government
3.Federal government	29	FALSE	3	1	2	5	Federal government
5.Local government	29	FALSE	3	1	2	5	Local government
5.Private company	29	TRUE	3	1	2	5	Private company
5.Self-employed	29	FALSE	3	1	2	5	Self-employed
5.State government	29	FALSE	3	1	2	5	State government

Basic unordered choice models

① Multinomial logit

② Conditional logit

- Explanatory variables are individual and alternative specific; they vary from individual to individual and are different for each choice the consumer might make.
- In conditional logit besides entities characteristics, we have to have alternatives' characteristics.
- For example choice of fishing mode: beach, pier, private boat, charter boat, while we have the price of each alternative mode, catch rate per alternative, **income**.

Let Y_i be random variable that indicates the choice made. McFadden (1974a) has shown that if (and only if) the J disturbances are independent and identically distributed with Gumbel (type 1 extreme value) distribution,

$$F(\varepsilon_{ij}) = \exp(-\exp(-\varepsilon_{ij})), \quad (4)$$

then

$$\text{Prob}(Y_i = j) = \frac{\exp(\mathbf{z}'_{ij}\boldsymbol{\theta})}{\sum_{j=1}^J \exp(\mathbf{z}'_{ij}\boldsymbol{\theta})}, \quad (5)$$

which leads to what is called the **conditional logit model**.

It is often labelled the multinomial logit model, but this wording conflicts with the usual name for the model discussed [earlier], which differs slightly. Although the distinction turns out to be purely artificial, we will maintain it for the present.

Attributes vs Characteristics

Greene p. 842:

Utility depends on z_{ij} , which includes aspects specific to the individual as well to the choices. It is useful to distinguish them. Let $z_{ij} = [x_{ij}, w_i]$ and partition θ conformably into $[\beta', \alpha']'$.

The components of x_{ij} are typically called the **attributes** of the choices. But w_i contains the **characteristics** of the individual and is, therefore, the same for all choices.

Cameron and Trivedi p. 493:

Some regressors, such as gender, do not vary across alternatives and are called case-specific or alternative-invariant regressors.

Other regressors, such as price, may vary across alternatives and are called alternative-specific or case-invariant regressors.

Attributes vs Characteristics

- <https://cran.r-project.org/web/packages/mlogit/vignettes/e1mlogit.html>
- The problem set uses data on choice of heating system in California houses. The data set Heating from the mlogit package contains the data in R format. The observations consist of single-family houses in California that were newly built and had central air-conditioning. The choice is among heating systems. Five types of systems are considered to have been possible:
 - gas central (gc),
 - gas room (gr),
 - electric central (ec),
 - electric room (er),
 - heat pump (hp).

There are 900 observations with the following variables:

- `idcase` gives the observation number (1-900),
- `depvar` identifies the chosen alternative (`gc`, `gr`, `ec`, `er`, `hp`),
- `ic.alt` is the installation cost for the 5 alternatives,
- `oc.alt` is the annual operating cost for the 5 alternatives,
- `income` is the annual income of the household,
- `agehed` is the age of the household head,
- `rooms` is the number of rooms in the house,
- `region` a factor with levels `ncost1` (northern coastal region), `scost1` (southern coastal region), `mountn` (mountain region), `valley` (central valley region).

Multinomial logit or conditional logit?

- ① Fishing mode (pier, beach, boat, charter); covariates: price for chosen alternative, catch rate for chosen alternative.

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- ③ Fishing mode (pier, beach, boat, charter); covariates: price for chosen alternative, catch rate for chosen alternative, income, age, year of education.

Multinomial logit or conditional logit?

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- ③ Fishing mode (pier, beach, boat, charter); covariates: price for chosen alternative, catch rate for chosen alternative, income, age, year of education.
- ④ Soda choice (pepsi, coke, 7up); covariates: income, age, sex.

Multinomial logit or conditional logit?

- ① Fishing mode (pier, beach, boat, charter); covariates: price for chosen alternative, catch rate for chosen alternative.
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- ③ Fishing mode (pier, beach, boat, charter); covariates: price for chosen alternative, catch rate for chosen alternative, income, age, year of education.
- ④ Soda choice (pepsi, coke, 7up); covariates: income, age, sex.
- ⑤ Car model choice (Ford, Fiat, Ferrari); covariates: mileage per gallon, engine size, trunk size.

Multinomial logit or conditional logit?

- 1 Fishing mode (pier, beach, boat, charter); covariates: price for chosen alternative, catch rate for chosen alternative.
- 2 Fishing mode (pier, beach, boat, charter); covariates: income, age, year of education.
- 3 Fishing mode (pier, beach, boat, charter); covariates: price for chosen alternative, catch rate for chosen alternative, income, age, year of education.
- 4 Soda choice (pepsi, coke, 7up); covariates: income, age, sex.
- 5 Car model choice (Ford, Fiat, Ferrari); covariates: mileage per gallon, engine size, trunk size.
- 6 Crime committed (robbery, theft, assault, drugs, murder); covariates: tattoo type.

Conditional logit – cola2 dataset

```
// The dataset consists of choices of 1822 consumers from US.  
// ID - id number of a customer  
// CHOICE - Coke, Pepsi, 7-Up  
// PRICE      price of 2 liter soda  
// FEATURE    = 1 featured item at the time of purchase  
// DISPLAY    = 1 if displayed at time of purchase
```

variable name variable label

id	customer id
pepsi	= 1 if pepsi chosen, otherwise = 0
sevenup	= 1 if 7up chosen, otherwise = 0
coke	= 1 if coke chosen, otherwise = 0
pr_pepsi	price of 2 liter bottle of pepsi
pr_7up	price of 2 liter bottle of 7up
pr_coke	price of 2 liter bottle of coke
feat_pepsi	= 1 if pepsi is a featured item at the time of purchase, otherwise = 0
feat_7up	= 1 if 7up is a featured item at the time of purchase, otherwise = 0
feat_coke	= 1 if coke is a featured item at the time of purchase, otherwise = 0
disp_pepsi	= 1 if pepsi is displayed at time of purchase, otherwise = 0
disp_7up	= 1 if 7up is displayed at time of purchase, otherwise = 0
disp_coke	= 1 if coke is displayed at time of purchase, otherwise = 0

Conditional logit – cola2 dataset

Obs: 1822

id customer id
choice = 1 if brand chosen
price price of 2 liter soda
feature = 1 featured item at the time of purchase
display = 1 if displayed at time of purchase

Data source: Dr. Ron Niedrich and Dr. Danny Weathers,
Marketing Department, Louisiana State University

Note: These data are in stacked form. There are 3 records per customer id.
The first record is for Pepsi, the 2nd for 7-Up and the 3rd for Coke

Variable	Obs	Mean	Std. Dev.	Min	Max
-----+-----					
id	5466	911.5	526.0141	1	1822
choice	5466	.3333333	.4714476	0	1
price	5466	1.185134	.3059794	.16	2.99
feature	5466	.5087816	.4999686	0	1
display	5466	.3635199	.4810567	0	1

Conditional logit – cola2 dataset

id	choice	price	feature	display
1	0	1.79	0	0
1	0	1.79	0	0
1	1	1.79	0	0
2	0	1.79	0	0
2	0	1.79	0	0
2	1	0.89	1	1
3	0	1.41	0	0
3	0	0.84	0	1
3	1	0.89	1	0
4	0	1.79	0	0
4	0	1.79	0	0
4	1	1.33	1	0

"Pure" conditional logit

Call:

```
mlogit(formula = soda ~ feat + disp + price | 0, data = cola2,  
        method = "nr", print.level = 0)
```

Frequencies of alternatives:

```
      coke      pepsi sevenup  
0.27991 0.34577 0.37431
```

Coefficients :

	Estimate	Std. Error	t-value	Pr(> t)
feat	-0.010604	0.079937	-0.1326	0.8945
disp	0.462448	0.093048	4.9700	6.696e-07 ***
price	-1.744453	0.179932	-9.6950	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -1822.2

"Pure" conditional logit marginal effects

```
# marginal effects
z <- with(coola2, data.frame(
  feat=apply(feat, index(mlogit1)$alt, mean),
  disp=apply(disp, index(mlogit1)$alt, mean),
  price=apply(price, index(mlogit1)$alt, mean)))
> z
```

	feat	disp	price
coke	0.4956092	0.3358946	1.210307
pepsi	0.5148189	0.3117453	1.227453
sevenup	0.5159166	0.4429199	1.117640


```
z[,1:2] = 1
```



```
> effects(mlogit1, covariate = "price", data = z)
```

	coke	pepsi	sevenup
coke	-0.3782432	0.1710623	0.2071809
pepsi	0.1710623	-0.3721380	0.2010757
sevenup	0.2071811	0.2010760	-0.4082570

"Pure" conditional logit marginal effects

```
# marginal effects
> effects(mlogit1, covariate = "disp", data = z)
```

	coke	pepsi	sevenup
coke	0.09944915	-0.04348699	-0.05596216
pepsi	-0.04348699	0.09718597	-0.05369898
sevenup	-0.05596214	-0.05369896	0.10966110


```
> effects(mlogit1, covariate = "feat", data = z)
```

	coke	pepsi	sevenup
coke	-0.0022803146	0.0009971329	0.001283182
pepsi	0.0009971329	-0.0022284210	0.001231288
sevenup	0.0012831817	0.0012312881	-0.002514470

"Pure" conditional logit – average marginal effects

```
cola3.0 = cola2
cola3.0$price[seq(from=1, to=1822, by=3)] = z[1,3]
cola3.0$price[seq(from=2, to=1822, by=3)] = z[2,3]
cola3.0$price[seq(from=3, to=1822, by=3)] = z[3,3]
cola3.0$feat=0
cola3.0$disp=0
pr0 = predict(mlogit1,newdata=cola3.0)
pr0 = fitted(mlogit1,newdata=cola3.0, type="probabilities")

cola3.1 = cola3.0
cola3.1$feat = 1
pr0 = fitted(mlogit1,newdata=cola3.1, type="probabilities")

# AME for disp
colMeans(pr1-pr0)
```

coke	pepsi	sevenup
0.004466432	0.007987079	-0.012453511

"Impure" conditional model

- A.C. Cameron, P.K. Trivedi, *Microeconometrics Using Stata*, p. 507-508
- When some or all regressors are alternative specific, the conditional logit model is used. The model specifies that

$$p_{ij} = \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta} + \mathbf{z}'_i\boldsymbol{\gamma}_j)}{\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta} + \mathbf{z}'_i\boldsymbol{\gamma}_l)} \quad j = 1, \dots, m \quad (6)$$

- where $\mathbf{x}_{ij}$ are alternative-specific regressors and $\mathbf{z}_i$ are case-specific regressors.
- To ensure identification, one of the $\boldsymbol{\gamma}_j$ is set to zero, as for MNL model.
- Some authors call the model above a mixed logit model.

Coefficient interpretation

- A.C. Cameron, P.K. Trivedi, *Microeconometrics Using Stata*, p. 507-508
- Coefficients of alternative-specific regressors are easily interpreted. The alternative-specific regressor can be denoted by x_r with the coefficient β_r . The effect of a change in x_{rik} , which is the value of x_r for individual i and alternative k , is

$$\frac{\partial p_{ij}}{\partial x_{rik}} = \begin{cases} p_{ij}(1 - p_{ij})\beta_r & j = k \\ -p_{ij}p_{ik}\beta_r & j \neq k \end{cases} \quad (7)$$

- If $\beta_r > 0$, then the own-effect is positive, and the cross-effect is negative.
- Here the negative price coefficient of -0.025 means that if the price of one mode of fishing increases, then demand for that mode decreases and demand for all other modes increases, as expected.
- For catch rate, the positive coefficient of 0.36 means a higher catch rate for one mode of fishing increases the demand for that mode and decreases the demand for the other modes.

"Impure" conditional logit

```
> mlogit1 = mlogit(mode ~ price+catch | income, data = Fish)
> summary(mlogit1)
```

Coefficients :

	Estimate	Std. Error	t-value	Pr(> t)	
boat:(intercept)	5.2728e-01	2.2279e-01	2.3667	0.0179485	*
charter:(intercept)	1.6944e+00	2.2405e-01	7.5624	3.952e-14	***
pier:(intercept)	7.7796e-01	2.2049e-01	3.5283	0.0004183	***
price	-2.5117e-02	1.7317e-03	-14.5042	< 2.2e-16	***
catch	3.5778e-01	1.0977e-01	3.2593	0.0011170	**
boat:income	8.9440e-05	5.0067e-05	1.7864	0.0740345	.
charter:income	-3.3292e-05	5.0341e-05	-0.6613	0.5084031	
pier:income	-1.2758e-04	5.0640e-05	-2.5193	0.0117582	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -1215.1

McFadden R²: 0.18868

Likelihood ratio test : chisq = 565.17 (p.value = < 2.22e-16)

"Impure" conditional logit – interpretation

```
> 1000*effects(mlogit1, covariate = "income",
+             data = z)
             beach          boat          charter          pier
-0.0007214174  0.0317613102 -0.0217339147 -0.0093059796
> effects(mlogit1, covariate = "price", data = z)
             beach          boat          charter          pier
beach  -1.249124e-03  0.0005531588  0.0006091541  8.681094e-05
boat    5.531588e-04 -0.0061167595  0.0048696270  6.939737e-04
charter 6.091542e-04  0.0048696271 -0.0062430047  7.642235e-04
pier    8.681094e-05  0.0006939736  0.0007642234 -1.545008e-03
> effects(mlogit1, covariate = "catch", data = z)
             beach          boat          charter          pier
beach    0.017793621 -0.007879681 -0.008677329 -0.001236612
boat    -0.007879671  0.087132394 -0.069367164 -0.009885559
charter -0.008677316 -0.069367154  0.088930726 -0.010886256
pier    -0.001236612 -0.009885571 -0.010886272  0.022008455
```

Unordered choice models in R – mlogit package

```
## a pure "conditional" model
```

```
summary(mlogit(mode ~ price + catch, data = Fish))
```

```
## a pure "multinomial model"
```

```
summary(mlogit(mode ~ 0 | income, data = Fish))
```

```
## which can also be estimated using multinom (package nnet)
```

```
library("nnet")
```

```
summary(multinom(mode ~ income, data = Fishing))
```

```
## a "mixed" model
```

```
m <- mlogit(mode ~ price + catch | income, data = Fish)
```

```
summary(m)
```

Prison tattoos

A statistical analysis of the art on convicts' bodies

What can be learned from a prisoner's tattoos



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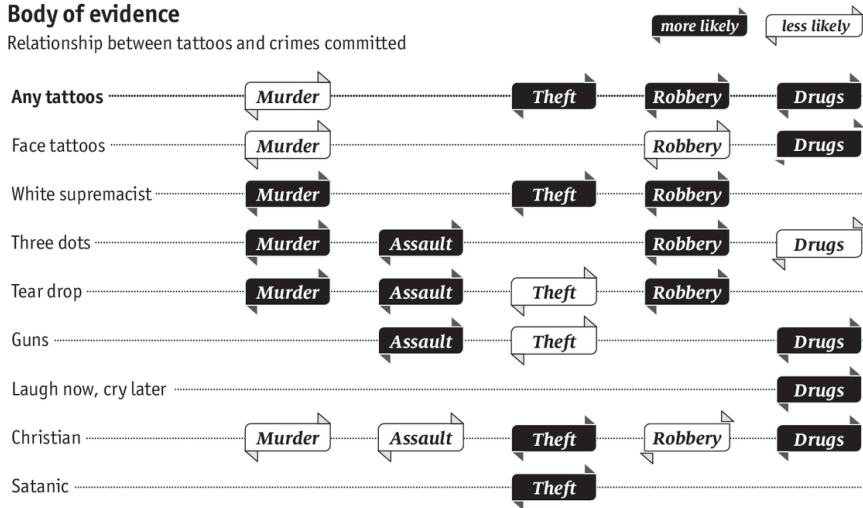
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Relationship between tattoos and crimes committed

Body of evidence

Relationship between tattoos and crimes committed



Sources: Florida Department of Corrections; *The Economist*